



OPEN WTAM-YOLO: a YOLOv11-based method for pulmonary nodule detection

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Lung cancer is among the malignancies with the highest incidence and mortality rates worldwide, and it poses a serious threat to human health. Increasing the accuracy of pulmonary nodule detection in CT images is essential for the early diagnosis and treatment of lung cancer. However, the grayscale characteristics of lung CT images, together with the variability in the sizes and morphologies of nodules, make the existing detection models prone to false positives and false negatives, posing challenges for achieving accurate detection. To address these problems, an improved WTAM-YOLO model based on YOLOv11 is proposed in this study. The model features four main improvements: a wavelet convolution approach to expand the receptive field, a lightweight convolutional block attention module (CBAM) to enhance the key feature representations, a hierarchical residual attention mixer (HRAMi) module to improve the multiscale detection performance of the model, and an improved exponential moving average (iEMA) module to strengthen the detail capture ability of the model and reduce the number of false positives. Experiments are conducted with a pulmonary nodule dataset acquired from the Roboflow platform and the LUNA16 dataset. Compared with those of YOLOv11, the proposed model improves the mAP@50 values by 3.4% and 2.5%, the mAP@75 values by 9.5% and 7.0%, the precision values by 4.4% and 0.7%, and the recall values by 2.3% and 4.6% based on the Roboflow and LUNA16 datasets, respectively.

Keywords Pulmonary nodule detection, YOLOv11 model, Wavelet convolution, Hierarchical complementary attention mixer

According to data from the World Health Organization, lung cancer is among the most common malignancies worldwide, and it is the leading cause of cancer-related death, with approximately 2 million new cases and nearly 1.8 million deaths each year¹. Pulmonary nodules are early manifestations of lung cancer. If lung cancer is detected and treated at an early stage, the associated five-year survival rate can exceed 80%. In contrast, for patients diagnosed at advanced stages without timely interventions, the five-year survival rate is often less than 20%². Therefore, the accurate and early detection of pulmonary nodules is crucial for improving patient survival rates and represents a key step toward lung cancer prevention. The traditional imaging-based assessment and preliminary localization of pulmonary nodules rely mainly on manual interpretation, where radiologists inspect chest CT images to determine the presence of nodules. However, nodules vary greatly in their shapes and sizes and often appear irregular³, making visual identification difficult. This challenge is further amplified in cases with small nodules or ground-glass nodules, which lack distinct features and are especially difficult to recognize with the naked eye⁴. In addition, radiologists working in high-intensity clinical environments for extended periods are prone to visual fatigue, which not only reduces their diagnostic efficiency but also increases the risks of false positives and missed detections⁵.

In recent years, deep learning has achieved substantial progress in medical image analysis. Among existing object detection algorithms, the YOLO series has attracted considerable attention in pulmonary nodule detection due to its high efficiency and real-time performance⁶. Nevertheless, several critical challenges remain in practical clinical applications. For images with low contrast and complex anatomical backgrounds, nodule features are often difficult to distinguish from adjacent tissues, leading to increased risks of false-positives and missed detections. Furthermore, pulmonary nodules exhibit substantial variability in terms of scale, morphology, and texture, and current methods remain limited in their cross-scale feature fusion and global semantic modeling abilities, making it difficult to effectively capture both local structural details and global contextual information. In addition, clinical diagnostic settings require detection systems to balance accuracy, speed, and stability;

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however, improvements in detection accuracy are often accompanied by increased computational costs and training instability, making it challenging to achieve an optimal trade-off between precision and efficiency.

These clinical and task-level challenges are technically reflected in the structural limitations of existing YOLO variants, particularly in fine-grained feature representation and multiscale fusion mechanisms. Although the YOLO family has continuously evolved for general object detection, achieving notable improvements in accuracy and efficiency through architectural refinements and enhanced feature pyramid designs, its capability to represent fine-grained structural details and perform effective cross-scale feature integration remains insufficient for pulmonary nodule detection. Recent studies have incorporated wavelet transforms or frequency-domain modeling into the YOLO framework; namely, the discrete wavelet transform (DWT) has been applied to decompose feature maps into multiple frequency sub-bands and enhance high-frequency edge and detail responses, or wavelet-based attention modules have been embedded within the backbone and feature fusion stages to increase sensitivity to texture and structural cues. However, these approaches typically treat frequency-domain modeling as a localized feature enhancement operator, restricting its influence to specific network layers or feature maps. Consequently, without systematic design of information flow pathways or functional role allocation, the propagation of frequency-domain information across hierarchical feature levels, its role in multiscale semantic organization, and its coordinated interaction with spatial-domain features remain largely dependent on the original YOLO architecture. From a methodological perspective, most existing wavelet-attention-based YOLO variants focus on module-level or operator-level refinements while lacking a holistic architectural strategy for hierarchical propagation, semantic organization, and task-driven scheduling of frequency-domain features. From a feature encoding standpoint, current YOLO models rely primarily on spatial-domain convolution for local region modeling, which constrains their ability to preserve high-frequency edge information and subtle texture variations. As a result, fine-grained density variations within target regions are prone to attenuation or oversmoothing in deeper network layers. During multiscale feature fusion, existing methods often emphasize scale alignment and feature concatenation, yet insufficient explicit constraints on semantic consistency and inter-level correlations may lead to intrinsic feature inconsistency, thereby weakening the overall discriminative capability. Moreover, commonly used attention mechanisms tend to prioritize global channel-wise or spatial reweighting while remaining less effective at highlighting locally critical regions, limiting the prominence of target-specific information within hierarchical feature representations.

To address these limitations, this study proposes an improved WTAM-YOLO model for pulmonary nodule detection, with an emphasis on a task-oriented network architecture integration strategy and a systematic cross-layer feature organization framework. Unlike prior YOLO-based approaches that introduce wavelet decomposition or frequency-domain attention primarily at local feature stages, the proposed model adopts a holistic hierarchical design that structurally orchestrates frequency-domain enhancement, multiscale feature organization, feature redistribution, and detection stability control across different network stages. This enables complementary feature representations across hierarchical levels, mitigating deep-feature oversmoothing, cross-scale semantic imbalance, and local response instability. Within this framework, the network architecture is restructured such that lower-level features prioritize edge and fine-detail encoding while intermediate and higher-level features reinforce semantic aggregation and discriminative representation. Furthermore, considering the characteristics of imaged pulmonary nodules in terms of scale distribution, grayscale contrast, and boundary morphology, the model introduces targeted optimization strategies for multiscale fusion pathways, feature scheduling mechanisms, and prediction stability. Compared with prior approaches that treat frequency-domain modeling as a local enhancement component, WTAM-YOLO emphasizes the systematic propagation of frequency-domain information across hierarchical feature streams and its coordinated integration with spatial-domain representations and establishes a stable and complementary feature hierarchy that improves the capability of the model to represent and detect pulmonary nodules across multiple scales. The main contributions of this work are as follows.

1. Multilevel feature enhancement and noise suppression: Wavelet convolution (WTConv) and the channel-spatial convolutional block attention module (CBAM) are incorporated into the backbone network to preserve fine-grained multiscale features while suppressing irrelevant redundant information. This design enables the highlighting of potential nodule regions under low-contrast and complex backgrounds, effectively alleviating the missed detection problem for cases involving tiny nodules.
2. Cross-scale feature interaction: A hierarchical complementary attention mixer (HRAMi) is incorporated into the feature fusion stage to dynamically aggregate the local details derived from different receptive fields with global contextual information, thereby achieving efficient cross-scale feature interaction. This module enhances the ability to capture nodule texture details at small scales while preserving structural consistency and global semantic representations at larger scales.
3. Small-scale detection enhancement and training stability: An iEMA module is integrated into the small-scale detection branch. By employing an exponential moving average mechanism, this module dynamically smooths and denoises fine-grained features, thereby strengthening the saliency of tiny nodules under complex backgrounds while mitigating gradient oscillations. As a result, the sensitivity of the small-object detection process is improved, and the stability of the model training procedure is enhanced.

Related work

The pulmonary nodule detection task has long faced challenges because of the small sizes of nodules, blurred boundaries, sample imbalance issues, and high computational costs. In this context, 2D, 3D, transformer-based, and multimodal methods have been successively introduced to gradually alleviate these difficulties.

In recent years, deep learning-based hybrid optimization approaches (e.g., integrating heuristic algorithms with integrated machine learning) have been increasingly adopted in medical data analysis⁷, especially in

medical image object detection and feature selection, yielding notable performance improvements⁸. Shaoqing Ren et al. first proposed the Faster R-CNN model, a 2D framework to generate candidate boxes and perform classification via a region proposal network (RPN), thus establishing an important baseline for early pulmonary nodule detection⁹. Although this model performs well for cases involving larger nodules, it has a high false-negative rate when detecting small nodules. To address this limitation, Li et al. introduced the multiscale Faster R-CNN, which incorporates a feature pyramid network (FPN) to fuse features across different scales, thereby enhancing the ability to detect small nodules¹⁰. Subsequently, Lin et al. developed the RetinaNet model, which integrates a focal loss to mitigate class imbalance issues in object detection tasks¹¹. To further improve the background suppression effect, Liu et al. proposed the attention Faster R-CNN, which employs an SE module to strengthen the attention mechanism of the model, reduce background interference, and focus more effectively on nodule regions. Additionally, Liu et al. presented a YOLOv7-based pulmonary nodule detection method that combines a deformable attention module with a WIoU loss function. The deformable attention module improves the sensitivity of the constructed model to small nodules, achieving superior performance relative to that of the original YOLOv7 network in complex backgrounds¹². The WIoU loss function adaptively adjusts the contributions of different regions to increase the resulting detection accuracy. Nevertheless, the method demands substantial computational resources, limiting its applicability in real-time detection scenarios. To overcome this challenge, Han et al. proposed BiRPN-YOLOvX, which integrates a bidirectional pyramid residual network (BiPRN) and leverages residual learning to improve the multiscale feature fusion process. The effective integration of multiscale information preserves low-level details while enhancing high-level semantic representations, allowing the model to more reliably distinguish nodules from background noise and improving its robustness in small-nodule detection tasks¹³. However, BiRPN-YOLOvX requires considerable computational resources and large training datasets, and its advantages are less pronounced in small-sample scenarios. Within convolution-based deep learning approaches, several researchers have proposed novel architectures, such as the TATHA model proposed by Banerjee et al.¹⁴, the CTDN model proposed by Singh D. P. et al.¹⁵, and TrionixNet¹⁶, to enhance medical image analysis performance. By improving feature extraction and information utilization strategies and effectively leveraging multiparameter imaging data, these models have substantially increased the accuracy and robustness of medical image analysis. Moreover, although these models perform well on 2D CT images, they rely solely on two-dimensional features and neglect spatial context information, which may reduce their detection accuracy when the boundaries of nodules with the surrounding tissues are unclear.

To address the limitations of 2D approaches in terms of modeling spatial context information, three-dimensional convolutional neural networks (3D CNNs) have gradually become the mainstream pulmonary nodule detection approaches, as they can fully exploit the volumetric information contained in CT images. Dou et al. first applied 3D convolution to pulmonary nodule detection¹⁷. Unlike 2D CNNs, which depend on single-slice features, 3D CNNs capture the spatial relationships between nodules and the surrounding tissues, significantly increasing the resulting detection accuracy, particularly for low-contrast or irregularly shaped nodules. By incorporating the Dice loss, Milletari et al. proposed a V-Net model to address the limited availability of annotations, which alleviated class imbalance issues and enhanced the stability achieved in small-sample training scenarios, thereby improving performance attained under scarce annotation conditions¹⁸. To further increase the computational efficiency of their model, Huang et al. developed 3D DenseNet, which employs dense connectivity to better reuse features and mitigate gradient vanishing, although this design increases the number of parameters and the computational overhead, imposing higher hardware requirements¹⁹. In contrast, DenseNet enhances interlayer feature propagation through dense skip connections, which substantially improves feature reuse and gradient flow. As a result, although it incurs increased computational cost, DenseNet delivers notable performance gains²⁰. Liu et al. introduced a lightweight 3D YOLO model using a CSPDarknet backbone, achieving a favorable tradeoff between speed and accuracy; however, it remains challenging for their model to adapt to small nodules and complex backgrounds²¹. Recent studies have also explored the application of U-Net and its variants to pulmonary nodule detection by leveraging encoder–decoder architectures to achieve multiscale feature fusion while incorporating convolutional or Inception modules to enhance local feature representation, thereby improving detection performance for small nodules and images with complex backgrounds²². Ma et al. proposed TiCNet, which incorporates transformer modules into a 3D CNN backbone, combining the local features extracted by convolutions with the global dependency modeling capabilities of transformers to enable efficient multiscale spatial information fusion²³.

The integration of transformer architectures has significantly advanced pulmonary nodule detection methods. Carion et al. first applied the vision transformer to object detection, capturing global context information through self-attention and enabling reliable long-range dependencies in complex scenarios²⁴. To address the integration of local and global features, Chen et al. developed 3D TransUNet, which combines CNNs for performing local feature extraction with transformers for conducting global context modeling, achieving effective pulmonary nodule segmentation and detection²⁵. Furthermore, Tang et al. proposed Swin-UNETR, which incorporates the hierarchical window attention mechanism of Swin transformers. This approach reduces the incurred computational complexity while establishing robust associations between local and global features, thereby enhancing the detection accuracy attained in challenging backgrounds²⁶. Nevertheless, transformer-based models exhibit limited generalizability under small-sample conditions and remain heavily reliant on large-scale annotated datasets. Similarly, Singh D. P. et al. proposed the Swin-T-McSWA framework, which integrates the hierarchical design of the Swin transformer with multicontext shifted window attention (McSWA) and lightweight T-blocks to enhance the capture of both local and global features in detection tasks while maintaining low computational overhead, making it suitable for automated detection systems in resource-constrained environments²⁷. However, although unsupervised learning methods can reduce reliance on large-scale annotated data, their performance under few-shot conditions and their ability to model complex features remain limited²⁸.

Meanwhile, although transformer-based models are effective at capturing global dependencies, they also require substantial labeled data and exhibit weaker generalization performance in small-sample settings²⁰.

With the incorporation of radiomics and clinical information, multimodal learning has increasingly become a key strategy in pulmonary nodule detection methods. Shen et al. proposed a multimodal CNN model that integrates CT images with the clinical data of patients. By employing feature-level joint learning, this model fully exploits the complementary information between imaging and clinical features, achieving improved detection accuracy²⁹. To enhance the cross-modal alignment capabilities of their model, Li et al. introduced a multimodal transformer, which captures intermodal relationships through self-attention and selectively fuses the key information. This approach attains improved detection performance and enhanced generalization, particularly in few-shot learning scenarios³⁰.

Despite the advances achieved by 3D convolutions, transformers, and multimodal methods, their clinical applicability remains limited. 3D CNNs effectively capture spatial context information but require substantial computational resources, restricting their use in large-scale 2D CT slice screening scenarios³¹. Transformer models exhibit improved discrimination in complex backgrounds via global dependency modeling, yet their high computational costs and training instability conflict with demands for efficient detection capabilities. Multimodal methods provide advantages in feature augmentation and few-shot learning tasks, but clinical data acquisition and modality alignment challenges constrain their practical implementations.

Given this background, the YOLO series, owing to its end-to-end one-stage detection paradigm and relatively low computational cost, demonstrates stronger applicability in terms of efficiently screening CT slices³². However, while YOLO models rely on local convolutional features, they often fail to adequately capture cross-scale information, leading to insufficient accuracy in low-contrast or small-nodule detection cases. To address this bottleneck, the Ultralytics team proposed YOLOv11 in 2024, introducing novel components such as a C3k2 block and C2PSA (a convolutional block with parallel spatial attention)³³. Specifically, YOLOv11 replaces some of the traditional convolutional structures with C3k2 modules, which split feature maps into two branches for undergoing separate convolutions before performing concatenation, thereby attaining an enhanced feature extraction ability. The spatial pyramid pooling-fast (SPPF) module is retained to optimize the computational efficiency of the model through multiscale pooling, supporting multisize object detection. In addition, the C2PSA attention mechanism is incorporated into the neck network, enabling the model to focus more on critical regions during the detection process while suppressing irrelevant background interference, thereby achieving improved detection accuracy³⁴. Compared with the previous YOLO variants, YOLOv11 demonstrates superior performance in complex scenarios and small-object detection cases. The architecture of YOLOv11 is illustrated in Fig. 1.

Although the YOLOv11 model has shown strong performance in general object detection scenarios, it still faces certain limitations with regard to the specific task of pulmonary nodule detection. Pulmonary nodules in CT images are typically small structures with diameters of only a few millimeters, and they exhibit considerable morphological variability³⁵, which places high demands on the fine-grained feature representation ability of the utilized model. However, the multiscale prediction mechanism of YOLOv11 may lose critical information

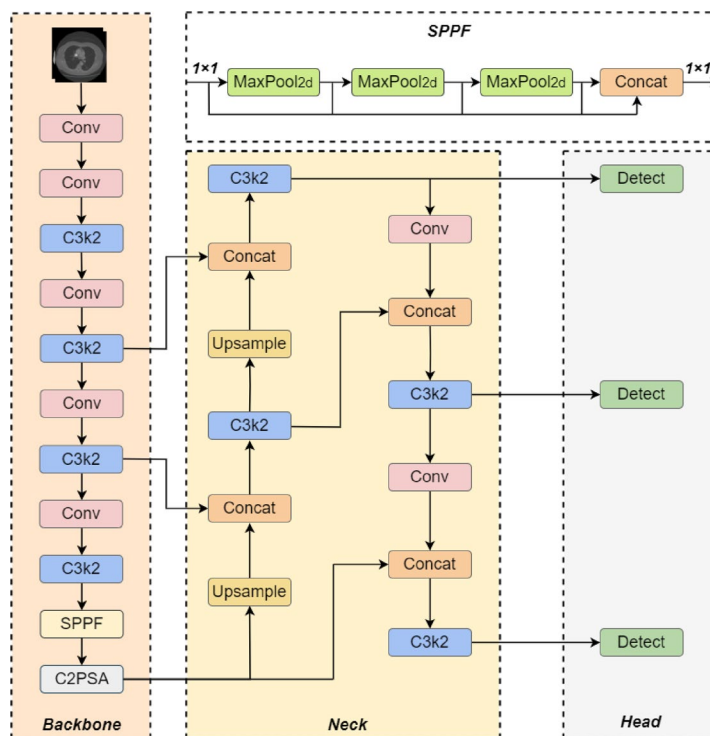


Fig. 1. Architecture of the YOLOv11 network.

because of the downsampling step performed when processing micronodules, thereby increasing the likelihood of missed detections. Moreover, its convolutional features rely heavily on local receptive fields and lack cross-scale and contextual constraints, making it difficult to distinguish nodules from similar structures such as blood vessels and bronchi, often resulting in false positives. In addition, the insufficient alignment of features across different levels during fusion further limits the sensitivity of the model to low-contrast nodules. Therefore, this study is focused on optimizing YOLOv11 in terms of feature preservation, cross-scale modeling, and semantic fusion to enhance its performance in pulmonary nodule detection tasks.

Methods

WTAM-YOLO model

To address the limitations of the YOLOv11 model in pulmonary nodule detection tasks, this study proposes an efficient detection framework, WTAM-YOLO, which is aimed at strengthening the ability to recognize small nodules and achieving improved overall detection performance. Although YOLOv11 has demonstrated notable detection accuracy and speed improvements over earlier versions in the YOLO series, it is still restricted by the limited receptive fields of convolutional kernels, which makes it difficult to sufficiently capture contextual information around tiny nodules and often results in missed detections or localization errors for nodules that are smaller than 5 mm in diameter. Therefore, the wavelet convolution (WTConv) is integrated with the C3k2 module in the backbone to form C3k2_WT, replacing the original C3k2. This design exploits the multiscale analysis capabilities of the wavelet transform to enhance multiscale feature representations and attain improved detection accuracy. In addition, within the improved YOLOv11 structure, the CBAM is placed after C2PSA and before the feature fusion process. At this stage, the feature maps already contain strong semantic information and serve as critical junctions for channel-spatial dependencies. Introducing the attention mechanism here suppresses shallow redundant features and provides more robust representations for the subsequent multiscale fusion operation, thereby ensuring both a theoretical justification and practical effectiveness. To further improve the ability of the developed model to detect nodules of varying sizes, three hierarchical complementary attention mixer (HRAMi) modules are incorporated into the key path of the neck, specifically after the upsampling and feature concatenation steps but before the C3k2 module. The modules are added after the second feature concatenation operation to optimize the computational efficiency of the model and avoid unnecessary overhead in the earlier stages. Furthermore, to alleviate the problem of missed detections when addressing tiny nodules, an exponential moving average (iEMA) module is introduced after the feature fusion and enhancement processes in the neck. By integrating features across multiple scales and semantic levels, this module substantially strengthens the detection robustness of the model in complex backgrounds and small-object scenarios. The architecture of WTAM-YOLO is illustrated in Fig. 2.

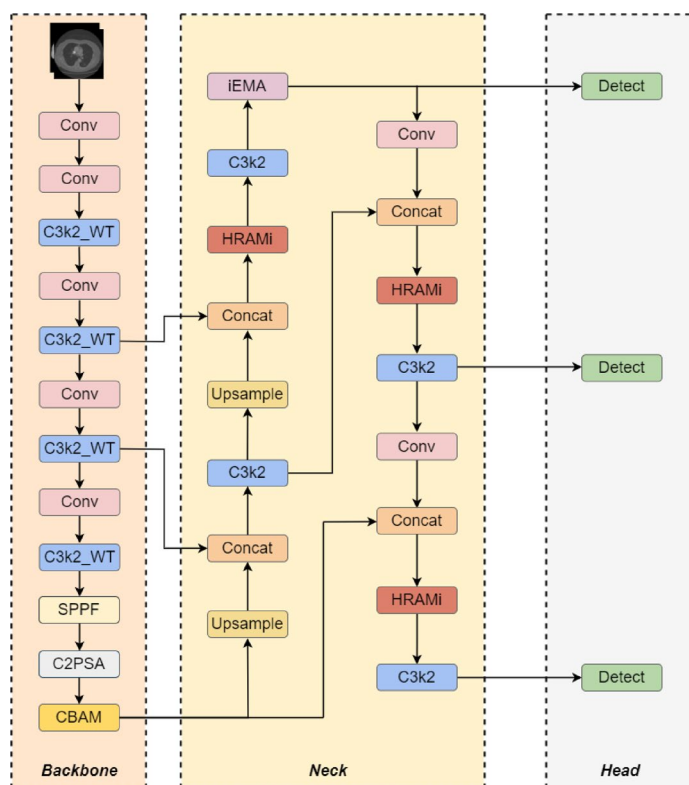


Fig. 2. WTAM-YOLO network architecture.

Wavelet convolution (WTConv)

In the YOLOv11 model, a single convolution operation is insufficient for effectively capturing information across multiple scales in images with complex backgrounds. To address this limitation, this study introduces the WTConv module, which leverages a two-dimensional Haar wavelet transform to perform multilevel decomposition on input images and thereby enhances the ability of the model to extract multiscale features. The two-dimensional Haar wavelet transform is adopted primarily because it can efficiently capture edge and contour information while maintaining low computational complexity during multiscale decomposition and can achieve a balanced representation of fine-grained local details and global structural characteristics. This property makes the WTConv module more sensitive to subtle pulmonary nodule structures, facilitating efficient image decomposition while achieving strong performance in terms of edge and contour representations³⁶. Specifically, the Haar wavelet transform uses four filters to decompose the input image into four subbands: a low-frequency component (LL), which captures global shapes and contours; a horizontal high-frequency component (LH), which focuses on horizontal edge information; a vertical high-frequency component (HL), which emphasizes vertical edge information; and a diagonal high-frequency component (HH), which preserves diagonal detail information. Through this multidimensional decomposition scheme, the module achieves fine-grained frequency separation at each level of the wavelet transform. After the convolution operations are completed, an inverse wavelet transform is applied to concatenate and fuse the convolution results of all the subbands, ultimately generating a complete output feature map. The architecture of WTConv is illustrated in Fig. 3.

As shown in Fig. 3, the input feature map X first undergoes a first wavelet transform (WT) and is decomposed into four subbands $X_{LL}^{(1)}$, $X_{LH}^{(1)}$, $X_{HL}^{(1)}$, and $X_{HH}^{(1)}$. Afterward, the low-frequency subband $X_{LL}^{(1)}$ undergoes a second wavelet transform, which further decomposes it into $X_{LL}^{(2)}$, $X_{LH}^{(2)}$, $X_{HL}^{(2)}$, and $X_{HH}^{(2)}$. These decomposed subbands are subsequently processed through convolution and scaling operations, and this is followed by reconstruction into a feature map via the inverse wavelet transform. The reconstructed feature map is then added to the feature map obtained from the baseline convolution operation to generate the output feature map of the WTConv module. The two-stage wavelet transform design enables the model to extract features at multiple scales. The first wavelet transform primarily captures global features, facilitating the recognition of the overall structures of pulmonary nodules, whereas the second wavelet transform refines the low-frequency subband to precisely capture local details, particularly the edges and subtle features of pulmonary nodules. Although the number of wavelet transform layers is increased, the computational efficiency of wavelet transforms ensures that this design improves the constructed feature representations without imposing a significant computational burden, thereby enhancing the performance of the model in pulmonary nodule detection cases. The computational formula for the output of the WTConv module is given in Eq. (1).

$$x_{out} = IWT(\sum_{i=0}^{L-1} wavelet_convs[i](W_{dec}^i(x))) + base_scale(base_conv(x)) \quad (1)$$

Here, x denotes the input feature map, IWT represents the inverse wavelet transform operation, L refers to the number of wavelet transform levels, $W_{dec}^i(x)$

denotes the wavelet convolution operation conducted at layer i , $base_conv$ indicates the basic convolution operation, and $base_scale$ refers to the basic scaling operation.

In the improved YOLOv11 network architecture, the WTConv module is incorporated into the C3k2 block of the backbone to replace the original convolutional branch, forming a C3k2_WT block. This module effectively addresses the limitations of conventional convolution operations in terms of capturing the multiscale features of pulmonary nodules. In this specific design, the convolution kernel is set to a size of 5×5 . This choice enlarges the receptive field while introducing richer contextual information, enabling the model to more effectively distinguish small nodules and their blurred boundaries under low-contrast and complex background conditions,

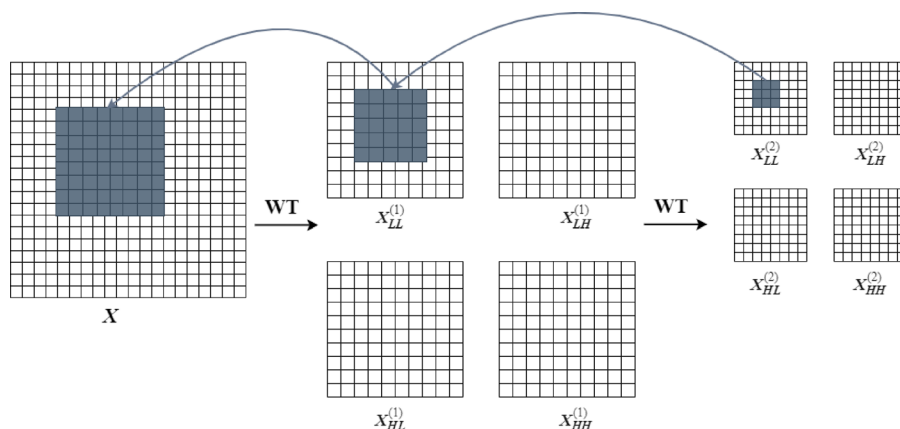


Fig. 3. WTConv module architecture.

thereby enhancing the overall representation capabilities of the model. Moreover, compared with larger convolution kernels (e.g., 7×7 or 9×9), the 5×5 kernel achieves a balance between receptive field expansion and computational cost, avoiding the training instability caused by redundant parameters. The original 3×3 convolutional branch is retained to ensure the sufficient extraction of local details while complementing the global contextual modeling procedure, thus allowing the network to achieve a balance between structural perception and fine-grained information. To ensure cross-layer feature transmission consistency, the input and output channels of the C3k2_WT block are strictly aligned with those of the backbone: when the feature dimensionality of the backbone is 256 channels, the input and output of the C3k2_WT block both possess 256 dimensions; in the deeper stages, the channel dimensions are maintained at 512 and 1024, respectively. This design guarantees the stability of the multiscale feature fusion process while preserving the lightweight advantage of the model and simultaneously enhancing its robustness. The architecture of the C3k2_WT block is illustrated in Fig. 4.

As shown in Fig. 4, the input features first undergo a convolution operation to adjust the channel dimensionality for the subsequent processing step. The features are then divided into two paths: one path proceeds directly to the feature fusion stage to preserve the original information, while the other is passed to the C3k_WT module. Within the C3k_WT module, the features are processed by the Bottleneck_WT block, where a 3×3 convolution extracts the basic features, followed by a 5×5 WTConv2d operation that performs multilevel decomposition. By applying the two-dimensional Haar wavelet transform, the WTConv2d module is able to capture multiscale information. The decomposed features are subsequently processed via convolution and fused through the inverse wavelet transform to generate a feature map enriched with multiscale information. After completing the feature fusion phase, the two paths are combined and passed through another convolution operation to produce the final output. This design replaces the standard convolution in the original C3k2 module with the WTConv2d module, thereby significantly enhancing the ability of the network to capture multiscale features.

CBAM

In lung nodule detection tasks, different channels respond differently to nodule features, with some emphasizing edge information and others focusing on shape characteristics. Additionally, the spatial locations of lung nodules in CT images are uncertain, and the surrounding anatomical structures can interfere with the ability to achieve accurate recognition. To address these challenges, this study incorporates the convolutional block attention module (CBAM), which combines channel and spatial attention mechanisms to enhance the feature representations along both dimensions³⁷. Along the channel axis, the CBAM dynamically evaluates the importance of each channel and assigns weights accordingly, highlighting the features that are relevant to lung nodules while suppressing less informative channels. This process strengthens the critical features and improves the recognition performance of the model. Along the spatial axis, the module captures the significance of each position in the feature map, concentrating on regions containing nodules while reducing the interference caused by the background and adjacent structures. This spatial focusing scheme improves the resulting localization accuracy and mitigates both false positives and false negatives. The integration of channel and spatial attention thus enables the precise localization and identification of lung nodules, effectively enhancing the detection accuracy and robustness of the model, particularly in complex CT images. The architecture of the CBAM is shown in Fig. 5.

As shown in Fig. 5, the CBAM first processes the input feature map through its channel attention submodule, and the resulting features are multiplied by the original input. The output is then passed to the spatial attention submodule, where it is again multiplied by the preceding features. Finally, the refined feature map is obtained as the output of the CBAM. The computation of the CBAM output is formulated as shown in Eq. (2).

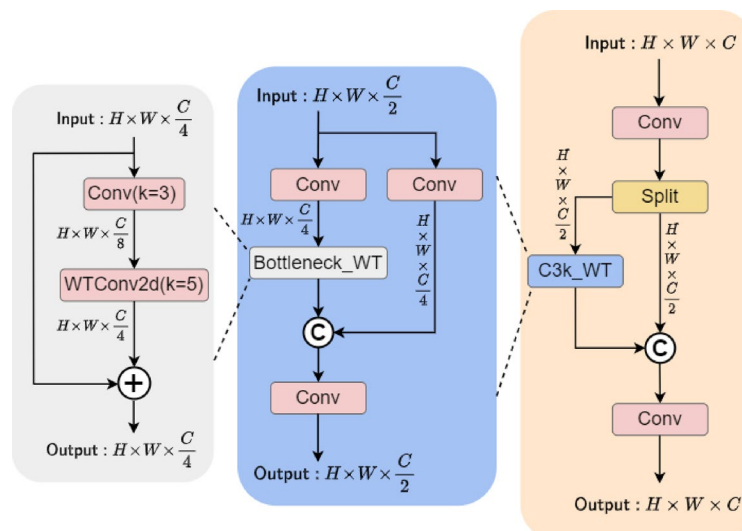


Fig. 4. C3k2_WT module architecture.

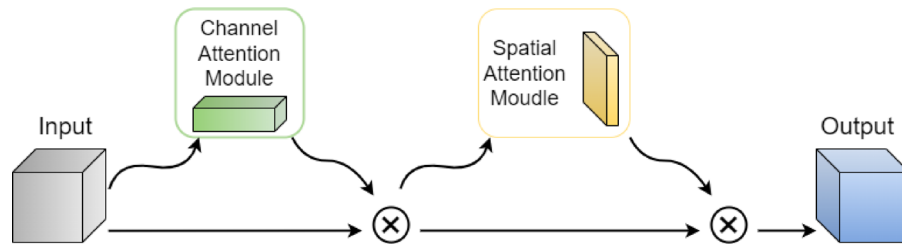


Fig. 5. CBAM architecture.

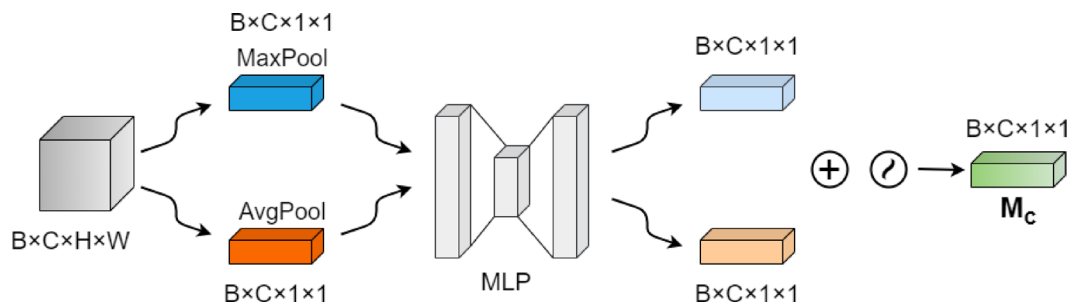


Fig. 6. Channel attention module architecture.

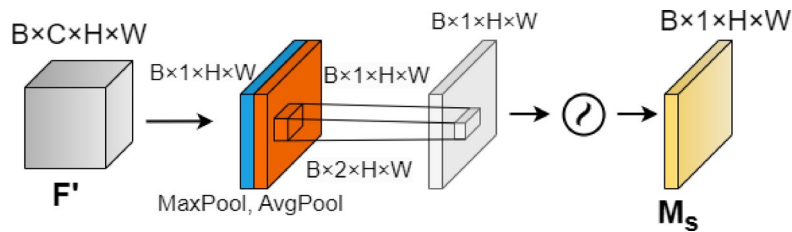


Fig. 7. Spatial attention module architecture.

$$F'' = M_s(M_c(F) \otimes F) \otimes F \quad (2)$$

where F denotes the input feature map, M_c represents the channel attention module, and M_s represents the spatial attention module.

The channel attention module receives a feature map with a shape of (B, C, H, W) , where B denotes the batch size, C is the number of channels, and H and W represent the height and width of the feature map, respectively. The input feature map is first processed through global average pooling and global maximum pooling to obtain two feature maps. These two feature maps are then fed into a multilayer perceptron (MLP) and passed through two 1×1 convolutional layers to adjust the channel dimensionality, producing two feature maps with shapes of $(B, C, 1, 1)$. Finally, the two feature maps are added and passed through a sigmoid activation function to generate the channel attention weight M_c . The architecture of the channel attention module is illustrated in Fig. 6.

The formula for calculating the channel attention weight is given in Eq. (3).

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \quad (3)$$

Similarly, the spatial attention module also receives a feature map with a shape of (B, C, H, W) . It computes the channelwise average and maximum values to generate two feature maps with sizes of $(B, 1, H, W)$. These two feature maps are then concatenated along the channel dimension to form a feature map with a shape of $(B, 2, H, W)$. A 7×7 convolution kernel is subsequently applied, producing a feature map with a size of $(B, 1, H, W)$. Finally, the sigmoid activation function is used to generate the spatial attention weight M_s . The architecture of the spatial attention module is illustrated in Fig. 7.

The formula for calculating the spatial attention weight is given in Eq. (4).

$$M_s(f) = \sigma(f^{7 \times 7}([AvgPool(F); MaxPool(F)])) \quad (4)$$

HRAMi module

In the hierarchical feature extraction processes of the traditional models, the loss of pixel-level information may lead to missed detections of tiny nodules and ground-glass nodules. Since the diameters of pulmonary nodules are usually less than 10 mm, the number of pixels a nodule occupies in a CT image is closely related to the resolution of the utilized CT equipment. For example, in common CT scans with resolutions of 0.5 mm per pixel, a 10-mm pulmonary nodule occupies approximately 20×20 pixels, whereas at 1 mm per pixel, it occupies approximately 10×10 pixels. After performing downsampling during feature extraction, the semantic information of the nodule almost disappears. To address this issue, this study introduces the HRAMi module, which integrates two variants: MobiVari1 and MobiVari2³⁸. Within the HRAMi module, MobiVari1 and MobiVari2 process the input feature maps in parallel, enabling the flexible adjustment of the feature extraction accuracy and efficiency of the model according to the given task requirements. Specifically, MobiVari1 reduces the computational complexity of the model, making it suitable for scenarios with limited computational resources and memory, where it can operate quickly and satisfy real-time demands; in contrast, MobiVari2 enhances feature representations through dilated and depthwise convolutions, making it better suited for detection tasks with complex backgrounds and variable object sizes and shapes, where it can achieve more precise localization and reduce the numbers of false negatives and false positives.

The HRAMi module receives multiscale feature maps as its inputs; therefore, in this study, it is incorporated at three different feature scales. Through multiscale feature fusion, HRAMi integrates information across these scales, allowing the model to leverage richer contextual cues when detecting tiny nodules, thereby improving its detection accuracy. In addition, the multidimensional attention mechanism of the model emphasizes salient features while suppressing irrelevant features, further enhancing the ability of the model to extract pulmonary nodule characteristics and reduce the number of missed detections. The network architecture of the HRAMi module is illustrated in Fig. 8.

The MobiVari1 module is designed on the basis of the MobileNetV1 architecture and adopts the depthwise-separable convolution strategy³⁹. Specifically, the module first applies a 3×3 convolution kernel independently to each input channel to perform depthwise convolution for spatial feature extraction purposes. A 1×1 pointwise convolution is subsequently employed across all the channels to adjust the interchannel correlations. When the numbers of input and output channels are identical, the module preserves the original input feature information by adding the original feature map to the processed feature map. In this way, the module reduces its computational resource requirements while retaining the information acquired from the original input feature map. The calculation formula for the output of the MobiVari1 module is given in Eq. (5).

$$X_{out} = \sigma(\text{Conv}_{pw}(\sigma(\text{conv}_{dw}(X)) + X)) \quad (5)$$

X denotes the original feature map, σ denotes the activation function, Conv_{dw} denotes the depthwise convolution, and Conv_{pw} denotes the pointwise convolution.

The MobiVari2 module is designed on the basis of the MobileNetV2 architecture. First, an expansion convolution with a kernel size of 1×1 is employed to increase the number of channels, thereby enhancing the feature representation ability of the model⁴⁰. Subsequently, a depthwise convolution with a kernel size of 3×3 is applied to the expanded feature map to strengthen the spatial feature extraction ability of the model. This is followed by a pointwise convolution with a kernel size of 1×1 for adjusting the interchannel correlations. When the numbers of input and output channels are the same, the original feature map is added to the processed feature map to preserve the original feature information. The output feature computation formula of the MobiVari2 module is given in Eq. (6).

$$X_{out} = \sigma(\text{Conv}_{pw}(\sigma(\text{Conv}_{dw}(\sigma(\text{Conv}_{exp}(X))) + \sigma(\text{Conv}_{exp}(X)))))) \quad (6)$$

X denotes the original feature map, σ denotes the activation function, Conv_{exp} denotes the expansion convolution, Conv_{dw} denotes the depthwise convolution, and Conv_{pw} denotes the pointwise convolution.

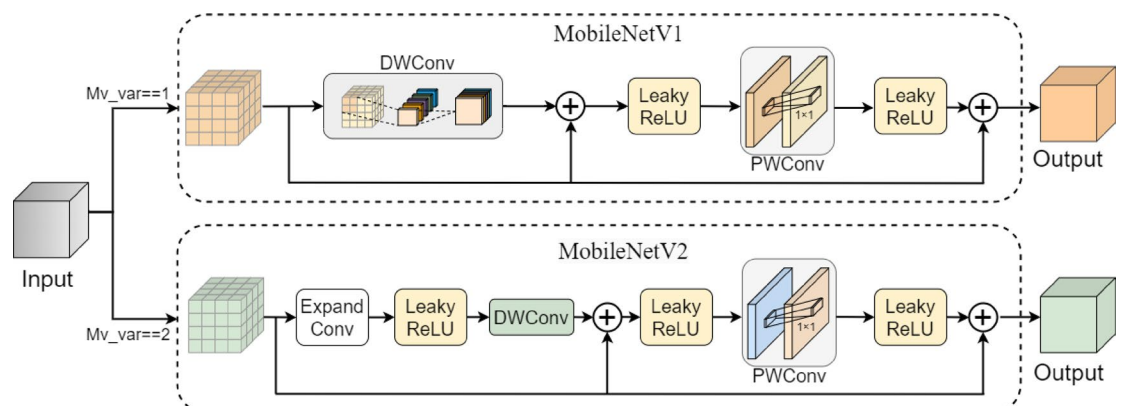


Fig. 8. HRAMi module architecture.

iEMA module

The traditional object detection models generally perform well in terms of recognizing large-scale targets; however, their performance is often unsatisfactory for cases involving pulmonary nodules, which vary in size and shape, particularly for tiny nodules. These tiny nodules occupy only a few pixels in the feature maps, making their features easily overwhelmed by background information, which often results in missed detections or false positives. To enhance the ability of the model to capture small objects, the iEMA module is incorporated into the neck network in this study. The iEMA module combines the advantages of the inverted residual block (iRMB) and the EMA module. The iRMB first expands the number of channels through a 1×1 convolution to strengthen the feature representations and then compresses the channels back to the original size via another 1×1 convolution⁴¹. The EMA module dynamically adjusts the weights of different channels and spatial locations to emphasize the critical features of the target. By integrating the inverted residual structure of the iRMB with the attention mechanism of the EMA module, the iEMA module effectively enhances the detection performance achieved for small objects⁴².

In this study, on the basis of the improved YOLOv11 network, the iEMA module is incorporated into the high-resolution feature map produced at the nineteenth layer of the neck network, which is a key layer for small object detection tasks because of its rich spatial details. The module first normalizes the feature map to stabilize the training process and then applies the EMA mechanism to combine channel and spatial attention, thereby highlighting the target features and suppressing background interference. To balance the computational efficiency of the model with its feature learning capacity, the grouping factor is set to 32, dividing the feature map channels into 32 groups to reduce the number of computations per group while preserving fine-grained details. The iEMA module employs two convolution operations: a 3×3 convolution to extract spatial features and enhance local details for detecting small objects, followed by a 1×1 convolution for adjusting the channel dimensions, ensuring consistency between the input and output. When the input and output dimensions match, the original feature map is added to the convolved output, which helps mitigate gradient vanishing issues and enables the retention of more original feature information. The processed features are then passed through a DropPath regularization process with a probability of 0.2, enhancing the generalizability of the model and preventing overfitting. Finally, when the dimensions are consistent, the module fuses the original and processed feature maps to generate the final output, preserving the integrity of small object features after multiple operations. The architecture of the iEMA module is illustrated in Fig. 9.

Loss function

In pulmonary nodule detection tasks, background or negative samples often outnumber the foreground or positive samples, leading to class imbalance issues that can adversely affect the detection performance of the utilized model. To address this issue, the focal loss function, which is effective for mitigating class imbalances, is employed in this study. The focal loss dynamically assigns different weights to easy-to-classify and hard-to-classify targets⁴³, enabling the model to focus more on challenging targets. This mechanism improves the ability of the model to detect difficult instances such as small nodules and ground-glass nodules. By using the focal loss function, the degree of imbalance between the foreground and background samples is effectively reduced. The computation of the focal loss function is presented in Eq. (7).

$$L_{Focal}(p, y) = -\alpha(1 - p)^{\gamma} y \log(p) - \alpha p^{\gamma}(1 - y) \log(1 - p) \quad (7)$$

Let p represent the predicted probability of the positive class and y represent the corresponding ground-truth label. When $y = 1$, the sample belongs to the positive class; when $y = 0$, the sample belongs to the negative class. The parameter α denotes the weight used to balance the positive and negative samples, which is generally set to 0.25 for positive samples and 0.75 for negative samples, primarily to address the class imbalance problem and to make the model pay more attention to positive samples. The parameter γ denotes the factor used to adjust the weights of the easy and hard samples; it is a nonnegative value and is usually set to 2.0 so that the model pays more attention to hard-to-classify samples.

Experiments

Experimental environment setup

The experimental environment and parameter configurations are shown in Table 1.

Datasets

Pulmonary nodule dataset derived from the Roboflow platform

The proposed model was evaluated using an open-source dataset obtained from the Roboflow platform. Roboflow provides more than 90,000 publicly available datasets and over 60 million images, which are curated and annotated by computer vision practitioners through Roboflow Universe⁴⁴. The dataset used in this study was

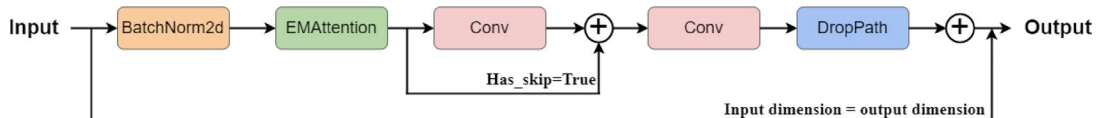


Fig. 9. Structure of the iEMA module.

Name	Parameter
Operating system	Ubuntu 24.04.1
Deep learning framework	PyTorch 2.7.0
Development environment	Python 3.9
CUDA version	CUDA 12.0
GPU	2*RTX 4090
GPU memory	2*24G
Image size	640 × 640
Random seed	42
Learning rate	0.0005
Learning rate decay factor	0.1
Batch size	16
Number of epochs	300
Early stopping patience	50

Table 1. Experimental environment and parameter configurations.

released in September 2024 and was designed specifically for pulmonary CT image object detection, offering valuable resources for research in this domain⁴⁵. It consists of annotated lung CT images, with both images and labels formatted in accordance with the YOLO standard to ensure consistency in terms of label structure and annotation format. From the perspective of medical annotation validity and labeling standardization, the dataset adheres to established medical conventions in terms of target class definition, bounding box annotation, and label formatting. In addition, it exhibits stable statistical characteristics with respect to object scale distribution, spatial coverage, and class proportion, without evident class imbalance or spatial bias. These properties ensure that the nodule annotations are semantically reliable, clearly categorized, and suitable for training and evaluating pulmonary nodule detection models. To further increase robustness to variations in imaging posture and scanning conditions and to improve training sample diversity and generalization capability, data augmentation was applied to the original images. While a fixed image resolution of 1024 × 1024 was maintained, the augmentation strategy included the following steps: each image had a 50% probability of horizontal flipping; subsequently, one of four rotation operations (no rotation, clockwise rotation, counterclockwise rotation, or 180° rotation) was applied with equal probability (25% each); finally, all the images were subjected to an additional random rotation within a range of − 15° to + 15°. The dataset contains a total of 2,757 lung CT images, of which 2,410 images were used for training, 230 images were used for validation, and 117 images were used for testing.

LUNA16

The Lung Nodule Analysis 16 (LUNA16) dataset was constructed based on the publicly available LIDC-IDRI database. It contains 888 CT scans and 1,186 annotated pulmonary nodules, which were labeled by four experienced radiologists as nonnodules, nodules smaller than 3 mm, and nodules with diameters greater than or equal to 3 mm⁴⁶. Because the original LUNA16 data consist of three-dimensional CT volumes, each volume was converted into two-dimensional slices along the axial plane to accommodate the proposed model. During slice selection, based on the nodule annotations, the central slice of each nodule along with one adjacent slice above and below were retained as positive samples. This strategy preserves the core morphological characteristics of nodules while incorporating contextual information from neighboring slices, thereby enhancing feature representation in two-dimensional convolutional networks. This multislice sampling approach partially mitigates the limitation of two-dimensional processing, that overlooks structural continuity, enabling the network to retain three-dimensional anatomical cues during two-dimensional feature learning and reducing potential adverse effects on clinical diagnostic reliability. To maintain a balanced ratio of positive and negative samples, negative samples were randomly selected from slices without nodules, allowing the network to learn both nodule-specific and background features. Because a single nodule may appear in multiple consecutive slices, samples corresponding to the same nodule were matched and merged based on their axial positions (Z-axis) and spatial distances in the two-dimensional plane to ensure that each nodule was uniquely represented in the dataset and prevent duplicates during counting.

After the two-dimensional slices were generated, each slice underwent preprocessing to emphasize lung parenchyma while suppressing background and surrounding tissue interference. All the images were converted to grayscale, and extracted features were normalized to ensure consistent value ranges across the samples. Based on the image characteristics and nodule distribution, an appropriate number of clustering centers (K) was selected; in this study, K was set to 2 to distinguish lung parenchyma from the background. Subsequently, the K-means clustering algorithm was employed to iteratively compute the distances from each pixel to the cluster centers, and each pixel was assigned to the nearest cluster⁴⁷. Based on the gray-level characteristics of the lung parenchyma, the cluster corresponding to the parenchymal region was determined. A binary mask was then generated according to the clustering results, where the lung parenchyma region was marked as the foreground (set to 1), and all other regions were marked as background areas (set to 0). Morphological opening was applied to remove small noise points, and morphological closing was used to fill small holes, ensuring the integrity of the target region⁴⁸. The connected components were subsequently identified, and since the lung parenchyma

typically corresponds to the largest connected region, this region was preserved, while the other regions were removed. The final segmentation mask was saved in the JPG format. After performing data augmentation, the dataset contained a total of 3,558 images, which were randomly divided into training, validation, and testing sets at a ratio of 8:1:1. The preprocessing pipeline of LUNA16 is illustrated in Fig. 10.

Evaluation metrics

To objectively evaluate the performance of the improved model in the lung nodule detection task, precision (P), recall (R), and mAP@50 (mean average precision at the 50% threshold) were used as evaluation metrics. In the detection task, lung nodules were regarded as positive samples, while other items were treated as negative samples. Precision measures the proportion of true-positive samples among all samples predicted as positive, whereas recall measures the proportion of actual positive samples that were correctly identified as positive by the model⁴⁹. The formulas for calculating precision (P) and recall (R) are given in Eqs. (8) and (9), respectively.

$$P = \frac{TP}{T + FP} \quad (8)$$

$$R = \frac{TP}{TP + FN} \quad (9)$$

In this study, TP (true positive) denotes the number of positive samples that were correctly predicted as positive by the model, FP (false positive) denotes the number of negative samples that were incorrectly predicted as positive, and FN (false negative) denotes the number of positive samples that were incorrectly predicted as negative.

The average precision (AP) was calculated as the area under the precision–recall curve to comprehensively evaluate the performance achieved by the model across different thresholds. For each class, precision and recall were first computed, and the AP was then obtained using an interpolation method. For mAP@50 and mAP@75, the mean AP was calculated by assuming that the IoU thresholds were fixed at 50% and 75%, respectively. The formulas for calculating AP, mAP@50, and mAP@75 are given in Eqs. (10), (11), and (12), respectively.

$$AP = \sum_{i=1}^{n-1} (r_{i+1} - r_i) \times \frac{p_i + p_{i+1}}{2} \quad (10)$$

$$mAP_{50} = \frac{1}{N} \sum_{i=1}^N AP_i^{50} \quad (11)$$

$$mAP_{75} = \frac{1}{N} \sum_{i=1}^N AP_i^{75} \quad (12)$$

where r_i denotes the recall at point i , p_i denotes the precision at point i , and AP_i denotes the average precision attained for class i .

Ablation study

In this study, ablation experiments were conducted on the lung nodule dataset acquired from the Roboflow platform and the LUNA16 dataset to evaluate the effects of different modules on the proposed model.

The experimental results obtained on the pulmonary nodule dataset provided by the Roboflow platform are presented in Table 2, where bold values indicate the best performance. As shown in the table, although the introduction of the WTConv module led to a slight decrease in mAP@50 of 0.1%, the precision, recall, and mAP@75 increased by 1.3%, 5.6%, and 4.2%, respectively, indicating that this module enhanced the ability to discriminate between positive and negative samples and strengthened the response to fine-grained nodule features through convolutional feature augmentation. The CBAM significantly enhanced the channel-wise and spatial feature representation of targets, causing the precision, recall, mAP@50, and mAP@75 to increase by 1.0%, 4.2%, 2.1%, and 5.2%, respectively, demonstrating that the attention mechanism can effectively reinforce the feature response of key regions and improve the accuracy of the model for detecting pulmonary nodules. The HRAMi module led to a 2.2% decrease in precision, but the recall, mAP@50, and mAP@75 still increased by 9.1%, 3.3%, and 7.4%, respectively, indicating that this module enhances the ability to detect small targets through multiscale high-resolution feature aggregation; however, the high-resolution details may interfere with the original features during cross-scale fusion, resulting in a slight decline in precision. The iEMA module

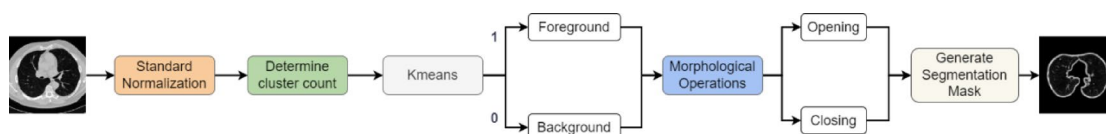


Fig. 10. Preprocessing pipeline employed for CT images.

WTConv	CBAM	HRAMi	iEMA	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
				90%	84.9%	91.6%	58.2%
√				91.3%	90.5%	91.5%	62.4%
	√			91%	89.1%	93.7%	63.4%
		√		87.8%	94%	94.9%	65.6%
			√	93.7%	88%	92.1%	62.7%
√	√			90.3%	85.5%	93.5%	63.7%
√		√		90%	82.9%	91.9%	60.2%
√			√	85.1%	87.2%	92.7%	63%
	√	√		89.6%	88.7%	88.3%	60.5%
	√		√	92%	87.2%	90.9%	62.6%
		√	√	88.2%	89.7%	93.6%	64.4%
√	√	√		91%	83.8%	93.9%	64.9%
√	√		√	89.9%	83.6%	91.7%	61.2%
√		√	√	88.4%	85.5%	92.5%	62.7%
	√	√	√	93.3%	84.6%	92.3%	62.8%
√	√	√	√	94.4%	87.2%	95%	67.7%

Table 2. Results of ablation experiments conducted on the lung nodule dataset derived from the Roboflow platform.

improved all the evaluated metrics relative to the baseline model, with the precision, recall, mAP@50, and mAP@75 increasing by 3.7%, 3.1%, 0.5%, and 4.5%, respectively, suggesting that this module stabilizes fine-grained target features via exponential moving average and thereby improves the robustness and consistency of the model for small-nodule detection. These results indicate that the incorporation of different modules impacts model performance to different degrees, validating their feasibility and effectiveness in the detection task.

Notably, using the CBAM alone yielded a recall of 89.1%, whereas using the HRAMi module alone yielded a recall of 94%. However, the combined use of both modules resulted in a 88.7% decrease in recall, indicating that the synergy did not preserve the advantage of using a single module in terms of recall. A possible explanation is that the CBAM reinforces local key regions in the input features via channel-wise and spatial attention, whereas the HRAMi module enhances small-target features through multiscale high-resolution feature fusion. When used together, the CBAM redundantly reweights the high-resolution small-target features provided by HRAMi, leading to the oversaturation of local feature responses, amplification of redundant information, and suppression of lower-resolution global contextual information. Such excessive local feature enhancement may cause uneven model responses across the overall nodule distribution, particularly when fine-grained nodules coexist with complex backgrounds, resulting in a partial weakening of some nodule features and a subsequent decrease in recall. In addition, the CBAM and the HRAMi module may focus on different spatial scales and channel distributions, with HRAMi emphasizing fine-grained information during multiscale fusion and the CBAM focusing on local key regions. This mismatch in scale alignment and attention distribution can prevent some high-resolution small-target features from being fully integrated, thereby reducing feature consistency and limiting recall improvement. Since mAP@50 and mAP@75 depend on both precision and recall, a decline in recall weakens the response of the model to certain nodule features, resulting in corresponding decreases in these metrics. This suggests that further optimization of the feature fusion strategy is necessary when modules are combined to balance the local attention and global contextual response.

Overall, compared with those of the original YOLOv11 baseline model, the precision, recall, mAP@50, and mAP@75 of the improved WTAM-YOLO model increased by 4.4%, 2.3%, 3.4%, and 9.5%, respectively, demonstrating that the enhanced model can more accurately localize and identify pulmonary nodules and improve overall detection performance.

The effects of different modules on the model were evaluated on the LUNA16 dataset, and the experimental results are presented in Table 3.

The bold values in Table 3 represent the optimal results. As shown in Table 3, the WTConv module caused a 1.8% decrease in recall but 1.5%, 0.9%, and 5.0% increases in precision, mAP@50, and mAP@75, respectively. The CBAM slightly reduced the precision by 1.3% while increasing the recall, mAP@50, and mAP@75 by 1.7%, 0.6%, and 3.3%, respectively. The HRAMi module led to minor reductions of 0.1% and 0.9% in precision and recall, respectively, yet the mAP@50 and mAP@75 increased by 1.8% and 1.5%, respectively. Notably, although the HRAMi module caused small decreases in precision and recall, its ability to preserve small-target features optimized the overall AP performance, ultimately resulting in significant gains in mAP@50 and mAP@75. The iEMA module improved the precision, recall, mAP@50, and mAP@75 by 1.1%, 0.2%, 1.2%, and 0.8%, respectively, demonstrating its effectiveness in enhancing the performance of the model for pulmonary nodule detection. The other results indicate that models integrating two or three modules consistently outperformed the baseline model. Overall, the improved model achieved 0.7%, 4.6%, 2.5%, and 7.0% increases in precision, recall, mAP@50, and mAP@75, respectively, confirming that the introduced modules enhance the overall performance of the model.

WTConv	CBAM	HRAmi	iEMA	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
				89.2%	86.9%	90.6%	72.5%
√				90.7%	85.1%	91.5%	77.5%
	√			87.9%	88.6%	91.2%	75.8%
		√		89.1%	86%	92.4%	74%
			√	90.3%	87.1%	91.8%	73.2%
√	√			88.5%	88.9%	91.3%	72.8%
√		√		90%	86.9%	91.7%	75.3%
√			√	89.1%	88.3%	90.3%	74.9%
	√	√		90.1%	86.6%	91.6%	74.6%
	√		√	89.4%	86.6%	91.3%	75.1%
		√	√	89.4%	87.4%	92.2%	74.8%
√	√	√		89.1%	86.6%	90.6%	75.7%
√	√		√	90%	86.6%	91.6%	73.4%
√		√	√	87.9%	91.1%	93.2%	78.3%
	√	√	√	91.4%	87.5%	92.4%	76.2%
√	√	√	√	89.9%	91.5%	93.1%	79.5%

Table 3. Results of ablation experiments conducted on the LUNA16 dataset.

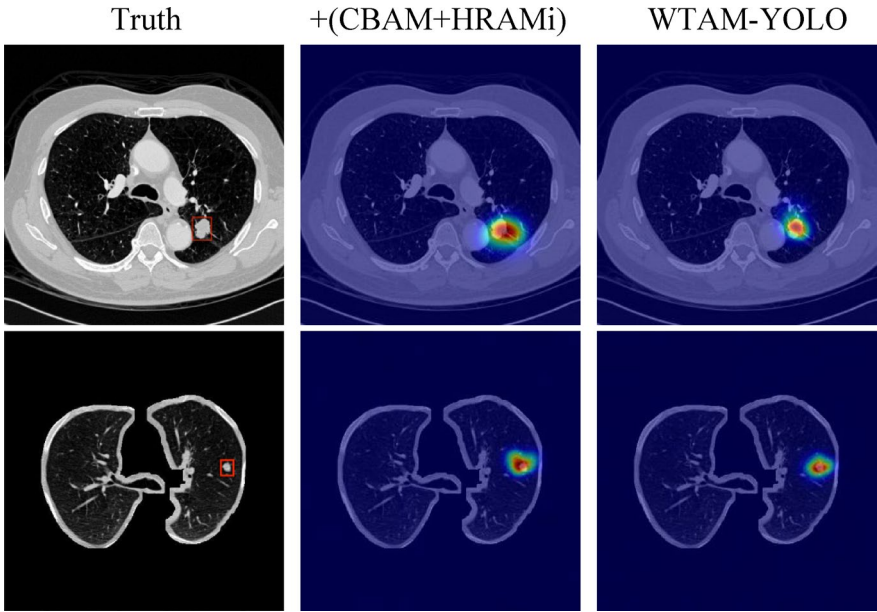


Fig. 11. Visual comparison of the attention response heatmaps for the improved modules.

To more intuitively demonstrate the impact of different module combinations on the internal attention distribution of the model, a visual comparison of the attention responses of the module combination model and the final optimized model on the lung nodule datasets from the Roboflow platform and LUNA16 is presented, as shown in Fig. 11.

As shown in Fig. 11, when the CBAM and the HRAmi module are jointly applied on both datasets, the model still successfully covers the true nodule regions. However, the attention responses exhibit a more dispersed spatial distribution, accompanied by slight shifts in the local activation centers. This phenomenon indicates that during the concurrent use of multiscale feature fusion and channel–spatial attention, different modules may compete in feature-weight allocation and partially redistribute responses that should focus on the lesion core to surrounding contextual structures. Therefore, this effect manifests as a reduction in localization and focus precision rather than the substantive missed detection of the nodules. Meanwhile, the optimized model consistently maintains a high degree of focus on the nodule core regions across both datasets, accurately captures critical features, and exhibits more consistent patterns in the attention distribution, thereby enabling precise localization and reliable detection of the lesion areas.

Based on the aforementioned multimodule ablation experiments, a systematic analysis was further conducted to assess the effect of the convolutional kernel size of the WTConv module on model performance. Kernel

Kernel size	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
1 × 1	92.1%	84.3%	90.4%	58.8%
3 × 3	89.2%	86.4%	91.8%	58%
5 × 5(Ours)	91.3%	90.5%	91.5%	62.4%
7 × 7	92.9%	84%	91.7%	56.7%
9 × 9	93.3%	85.2%	90.2%	58.2%

Table 4. Results of the ablation experiments on the Roboflow lung nodule dataset with different WTConv kernel sizes.

Kernel size	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
1 × 1	91%	81.6%	90.8%	73.6%
3 × 3	88.5%	83.8%	92.2%	76.9%
5 × 5(Ours)	90.7%	85.1%	91.5%	77.5%
7 × 7	90.2%	79.6%	91.5%	72.8%
9 × 9	92.1%	80.5%	90.6%	73.3%

Table 5. Results of the ablation experiments on the LUNA16 dataset with different WTConv kernel sizes.

Model	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
YOLOv11	90%	84.9%	91.6%	58.2%
+CBAM(Backbone)	91%	89.1%	93.7%	63.4%
+CBAM(Neck)	90.6%	88.6%	93.1%	62.5%
+CBAM(Head)	90.5%	88.3%	92.7%	61.9%

Table 6. Results of the CBAM position ablation experiments conducted on the lung nodule dataset derived from the Roboflow platform.

size ablation experiments were performed on both the pulmonary nodule dataset from the Roboflow platform and the LUNA16 dataset to evaluate the impact of different kernel dimensions. The experimental results are presented in Tables 4 and 5, with the bold values indicating the optimal results.

As shown in Tables 4 and 5, although smaller convolution kernels (e.g., 1 × 1 and 3 × 3) exhibited certain advantages in terms of some precision or mAP@50 metrics, their limited receptive fields constrained the ability to model contextual information, resulting in suboptimal recall and mAP@75 on both datasets. This indicates that small kernels remain insufficient for capturing complex nodule boundary structures and fine localization features. In contrast, larger convolution kernels (e.g., 7 × 7 and 9 × 9) expand the feature receptive field; however, the experimental results show that stable improvements in recall and mAP@75 are not achieved, suggesting that excessively large receptive fields may introduce redundant background responses and reduce the focus on local discriminative features. Among these kernels, the 5 × 5 convolution kernel achieved the highest recall and mAP@75 on the Roboflow dataset and maintained leading or near-optimal recall and mAP@75 on the LUNA16 dataset, demonstrating a better balance between contextual awareness and local feature discrimination. Therefore, considering cross-dataset stability and fine localization performance, a 5 × 5 kernel was adopted for the WTConv module.

To systematically evaluate the impact of the CBAM at different network positions, ablation experiments were conducted on the Roboflow lung nodule dataset and the LUNA16 dataset by varying the embedding positions of the CBAM to assess its role across different feature hierarchies. The experimental results are presented in Tables 6 and 7, with the bold values indicating the optimal results.

As shown in Tables 6 and 7, integrating the CBAM at the backbone stage yielded optimal performance across all metrics on the lung nodule dataset derived from the Roboflow platform. On the LUNA16 dataset, the precision at this position was slightly lower than that observed when the module was placed at the neck or head, which may be attributed to the global feature enhancement at the backbone stage introducing some background responses and thereby increasing the number of false-positives. However, the recall, mAP@50, and mAP@75 were all the highest, indicating that placing the CBAM at the backbone stage facilitates improved overall recognition and localization of pulmonary nodules. Therefore, the CBAM was integrated at the backbone in this study.

Similarly, ablation experiments were conducted on the grouping factor of the iEMA module to assess the impact of different grouping factors on detection performance. The results are presented in Tables 8 and 9, with the bold values indicating the optimal results.

As shown in Tables 8 and 9, smaller grouping factors (e.g., 4 or 8) yielded suboptimal performance across all metrics, likely due to excessively fine channel partitioning that dispersed information flow and reduced the

Model	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
YOLOv11	89.2%	86.9%	90.6%	72.5%
+CBAM(Backbone)	87.9%	88.6%	91.2%	75.8%
+CBAM(Neck)	88.1%	87.8%	91%	74.9%
+CBAM(Head)	87.7%	87.2%	90.8%	75.1%

Table 7. Results of the CBAM position ablation experiments conducted on the LUNA16 dataset.

Group factor	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
4	88%	87.6%	91.5%	56.7%
8	92.4%	83.1%	92.4%	55.1%
16	89.7%	88.5%	92.8%	60.7%
32(Ours)	93.7%	88%	92.1%	62.7%
64	91.1%	87.2%	93%	57.7%

Table 8. Results of the ablation experiments on the grouping factor of the iEMA module conducted on the lung nodule dataset derived from the Roboflow platform.

Group factor	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%
4	86.7%	85.4%	90.6%	68.4%
8	90.1%	85.3%	91.2%	70.5%
16	88.7%	87%	92.3%	72.7%
32(Ours)	90.3%	87.1%	91.8%	73.2%
64	89.1%	86%	92.7%	69.9%

Table 9. Results of the ablation experiments on the grouping factor of the iEMA module conducted on the LUNA16 dataset.

efficiency of overall feature utilization. When the grouping factor was set to 16 or 32, the recall and mAP@50 were comparable; however, both the precision and mAP@75 were lower for a factor of 16 than for a factor of 32, indicating certain limitations in feature integration, false-positive control, and fine-grained feature enhancement. Although a larger grouping factor of 64 slightly improved the mAP@50, precision, recall, and all the mAP@75 values decreased, suggesting that overly large groupings can weaken the fine-grained representation of local features and increase the risk of information loss after feature separation, thereby limiting accurate identification and the localization of small pulmonary nodules. In contrast, a grouping factor of 32 achieved relatively high precision, recall, and mAP@75 across both datasets, demonstrating superior overall performance and a good balance between local feature enhancement and global information integration. Therefore, a grouping factor of 32 was selected for the iEMA module to optimize both detection accuracy and feature representation capability.

Comparative experiments

The WTAM-YOLO model proposed in this study enhances both the accuracy of lung nodule detection and the overall performance of the model while maintaining a comparable number of parameters and a similar computational cost. To comprehensively assess the effectiveness of the improved model, we conducted experiments on lung nodule datasets derived from the Roboflow platform and LUNA16 and compared WTAM-YOLO with several mainstream detection models. The compared models included YOLO-Bi_FPN, which uses a bidirectional feature pyramid (Bi-FPN) to integrate features in both the bottom-up and top-down directions and combines weighted feature fusion, upsampling, downsampling, and convolutional operations⁵⁰; YOLO-SENetV2, which incorporates the SENetV2 attention mechanism along with multibranch and densely connected layers⁵¹; YOLO-SwinTransformer, which replaces the traditional convolutional backbone with a Swin transformer, improving feature representations via hierarchical feature extraction and window-based attention⁵²; YOLO-MobileNetV4, which employs MobileNetV4 as its backbone and leverages lightweight and depthwise-separable convolutions for efficiently extracting features⁵³; and YOLO-EfficientNetV2, which integrates the compound scaling, blocked gated convolution, and multiscale feature fusion modules of EfficientNetV2⁵⁴. The models were evaluated using precision, recall, mAP@50, the parameter count, and the number of floating-point operations (FLOPs).

First, the improved model was compared with the state-of-the-art detection models on the lung nodule dataset provided by the Roboflow platform, and the comparative results are summarized in Table 10, where the bold values denote the best performance.

As shown in Table 10, the proposed WTAM-YOLO achieved the best performance in terms of both mAP@50 and mAP@75, and its overall performance was significantly superior to that of the other models. Specifically,

although YOLO-SENetV2 exhibited slightly higher precision than WTAM-YOLO did, its recall, mAP@50, and mAP@75 were lower, with its precision being 0.8% higher, but its recall, mAP@50, and mAP@75 were 5.1%, 3.4%, and 9.2% lower, respectively. The YOLOv6 model achieved the highest recall; however, compared with those of WTAM-YOLO, its precision, mAP@50, and mAP@75 were 6.4%, 3.4%, and 7.2% lower, respectively, and its parameter and FLOP counts were substantially higher, reaching 2.0 and 2.1 times those of WTAM-YOLO. Similarly, although YOLOv5 achieved a precision of 95%, which was 0.6% higher than the 94.4% precision of WTAM-YOLO, its recall was only 81.2%, which is significantly lower than the 87.2% precision of WTAM-YOLO, and its mAP@50 and mAP@75 were 93.3% and 63.8%, which are both lower than the mAP@50 and mAP@75 (95.0% and 67.7%, respectively) of WTAM-YOLO. Furthermore, YOLOv5 required 9.12 M parameters and 24 G FLOPs, which are 3.26 and 3.16 times greater than those of WTAM-YOLO, respectively. These results indicate that although YOLOv5 had a slight advantage in terms of precision, WTAM-YOLO demonstrated superior overall performance and higher efficiency in terms of recall, mAP@50, mAP@75, and computational complexity. Notably, as a subsequent version of YOLOv11, YOLOv12 introduced deeper cross-layer feature fusion, dynamic label assignment, and an improved IoU loss function to optimize feature representation, sample matching, and bounding-box regression; however, in lung nodule detection, where lesions often exhibit low contrast and blurred boundaries and are difficult to distinguish from surrounding tissues, these optimizations enhanced high-level semantic features at the expense of detailed feature representations. In addition, the stricter sample matching mechanism decreased the recall to 77.8%, which is substantially lower than the 84.9% recall of YOLOv11, and its mAP@50 and mAP@75 were 89.6% and 53.1%, which are both lower than the mAP@50 and mAP@75 (91.6% and 58.2%, respectively) of YOLOv11. These findings demonstrate that cross-version upgrades of detection models do not necessarily lead to performance improvements in medical imaging tasks; rather, the adaptability of a model to the specific task is the key determinant.

To further validate the broad applicability of the proposed model, experiments were conducted on the LUNA16 dataset, and the comparative results are presented in Table 11, where the bold values indicate the best performance.

As shown in Table 11, the proposed WTAM-YOLO model demonstrated superior overall performance in the detection task. Compared with the other models, WTAM-YOLO achieved outstanding results across various key metrics, reaching an mAP@50 of 93.1% and an mAP@75 of 79.5%, both of which are significantly greater than those of the other models, and a recall of 91.5%, indicating excellent detection capability. In addition, WTAM-YOLO maintained a favorable balance between parameter count and computational cost, with 2.801 M parameters and 7.6 G FLOPs. This balance allowed the model to achieve high detection accuracy while reducing computational complexity and improving runtime efficiency, providing a solid foundation for efficient deployment in practical application scenarios. Although the precision was slightly lower than that of YOLOv6, the advantages of WTAM-YOLO in terms of recall, mAP@50, mAP@75, and computational efficiency make it a more favorable choice.

In summary, the experimental comparisons conducted on both datasets demonstrated that the WTAM-YOLO model achieved higher overall performance and detection accuracy than the other models did, indicating that the proposed model possesses both generalizability and robustness in object detection tasks. As shown in Tables 10 and 11, the proposed model introduces a slight increase in the number of parameters and computational cost. However, this increase remains within a controllable range and leads to substantial performance improvements. In particular, the model significantly enhances localization accuracy, improving the ability to determine the spatial position of pulmonary nodules and enabling precise identification of lesion boundaries, which is of considerable clinical value. From the perspective of the trade-off between efficiency and performance, the proposed model achieves a notable gain in detection performance at a modest increase in computational overhead. Compared with mainstream model scaling strategies, it attains superior performance

Method	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%	Params/M	FLOPs/G
YOLOv12	93.3%	77.8%	89.6%	53.1%	2.56	6.5
YOLOv11	90%	84.9%	91.6%	58.2%	2.62	6.6
YOLOv10	89%	82.1%	90.6%	57.1%	2.7	8.4
YOLOv8	84%	85%	91.6%	58.7%	3.96	10.3
YOLOv6	88%	88%	91.6%	60.5%	5.58	15.7
YOLOv5	95%	81.2%	93.3%	63.8%	9.12	24
YOLO-Bi_FPN	90.1%	81.2%	91.3%	51.8%	2.61	7.1
YOLO-SENetV2	95.2%	82.1%	91.6%	58.5%	2.54	6.4
YOLO-SwinTransformer	84.1%	76.6%	86.7%	52.4%	2.51	6.1
YOLO-MobileNetV4	89.7%	76.1%	86.5%	57.9%	2.17	5.3
YOLO-EfficientNetV2	90.2%	77.8%	88.9%	53.5%	2.09	5.3
Ours	94.4%	87.2%	95%	67.7%	2.801	7.6

Table 10. Comparative experimental results produced by different models on the lung nodule dataset derived from the Roboflow platform.

Method	Precision/%	Recall/%	mAP ₅₀ /%	mAP ₇₅ /%	Params/M	FLOPs/G
YOLOv12	83.8%	87.4%	88.9%	71.8%	2.56	6.5
YOLOv11	89.2%	86.9%	90.6%	72.5%	2.62	6.6
YOLOv10	90%	87.5%	92.4%	76.8%	2.7	8.4
YOLOv8	88.2%	88.3%	91.1%	76.1%	3.96	10.3
YOLOv6	92%	85.7%	93%	64.4%	5.58	15.7
YOLOv5	89.2%	88.6%	91.8%	78.3%	9.12	24
YOLO-Bi_FPN	89.3%	84.9%	90.6%	73.8%	2.61	7.1
YOLO-SENetV2	86.6%	86.6%	90.3%	73%	2.54	6.4
YOLO-SwinTransformer	82.8%	81.7%	86.9%	67.3%	2.51	6.1
YOLO-MobileNetV4	85.6%	84.9%	88.3%	71.5%	2.17	5.3
YOLO-EfficientNetV2	87%	88.3%	91%	76.7%	2.09	5.3
Ours	89.9%	91.5%	93.1%	79.5%	2.801	7.6

Table 11. Comparative results produced by different models on the LUNA16 dataset.

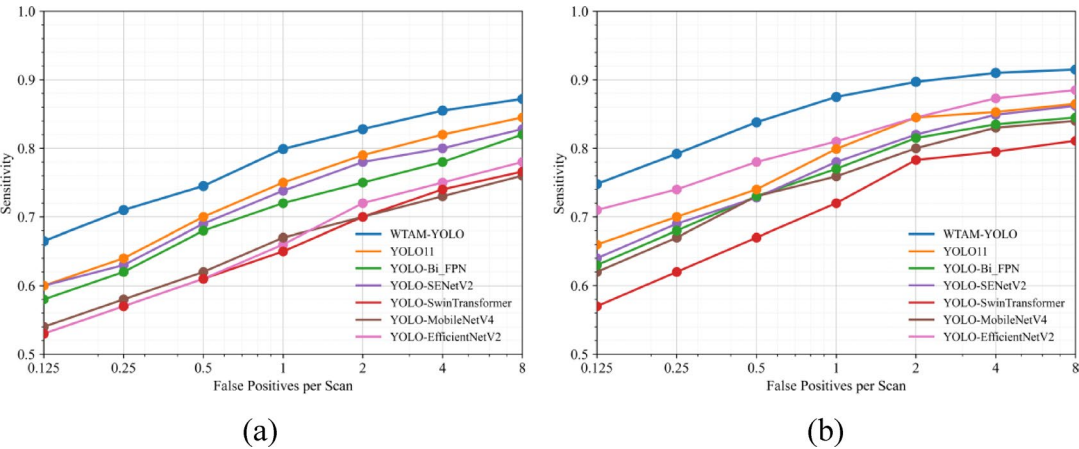


Fig. 12. FROC curves of the WTAM-YOLO model on the two datasets.

gains while maintaining relatively controllable increases in parameters and computation, thereby offering a feasible technical pathway for clinical deployment.

To further evaluate the detection performance of WTAM-YOLO on the datasets used in this study, we plotted the free-response receiver operating characteristic (FROC) curves. As shown in Fig. 12, (a) corresponds to the lung nodule dataset from the Roboflow platform, and (b) corresponds to the LUNA16 dataset. The horizontal axis represents the average number of false-positives per scan (FPs/scan), and the vertical axis represents sensitivity, illustrating the trade-off between sensitivity and the number of false-positives.

As shown by the FROC curves in Fig. 12, the WTAM-YOLO model outperform the other methods on both datasets, demonstrating higher greater sensitivity and fewer false-positives and indicating that the model achieves superior detection performance across different false-positives.

Based on the FROC analysis results, a visual comparison between the WTAM-YOLO model and several other models was conducted. Utilizing the lung nodule dataset acquired from the Roboflow platform and the LUNA16 dataset, the lung nodule detection capabilities of different models for were evaluated. The prediction results are illustrated in Figs. 13 and 14, where Fig. 13 shows the detection results produced by different models on the Roboflow lung nodule dataset, and Fig. 14 presents the detection results obtained on the LUNA16 dataset. Each row corresponds to one CT image, and each column corresponds to the prediction results of a specific model. Red circles indicate the positions of false detections induced by the models.

As shown in Figs. 13 and 14, the WTAM-YOLO model demonstrated higher confidence in detecting tiny nodules with diameters of less than 5 mm. Regarding the Roboflow lung nodule dataset, the first image in Fig. 13 shows that WTAM-YOLO achieved a confidence score of 0.84, compared with 0.77 for YOLO-Bi_FPN, while YOLO-EfficientNetV2 and YOLO-SwinTransformer produced false detections. According to the second image of Fig. 13, WTAM-YOLO yielded a confidence score of 0.80, whereas the other models all exhibited false detections, indicating that the proposed model achieved higher accuracy in terms of distinguishing small nodules from background noise. On the LUNA16 dataset, WTAM-YOLO also demonstrated superior performance with respect to detecting tiny nodules. For example, in the first image of Fig. 14, WTAM-YOLO

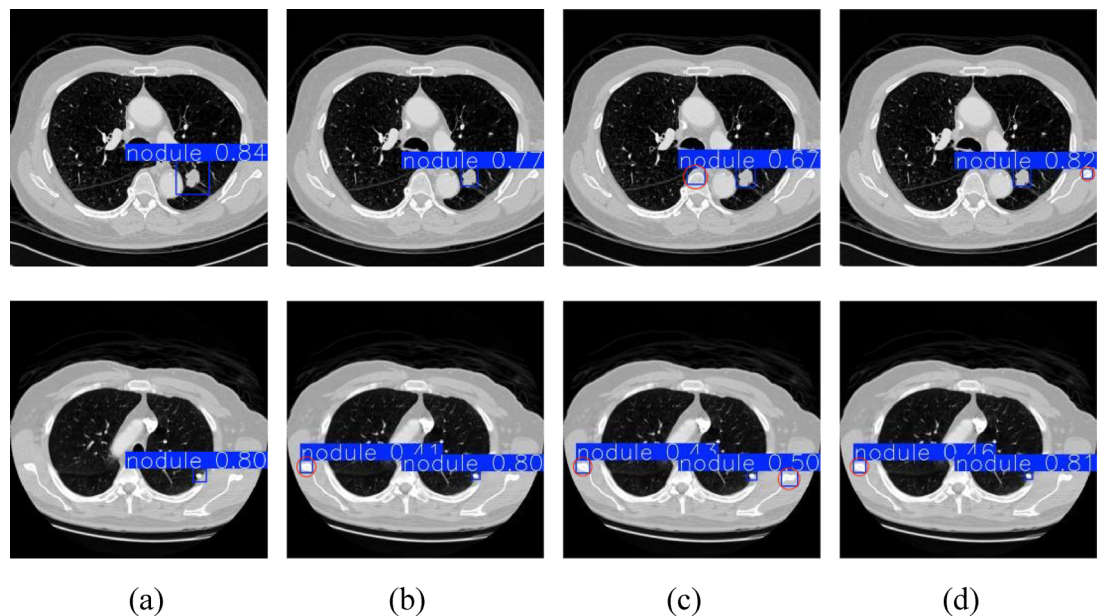


Fig. 13. Prediction results produced by different models on the Roboflow lung nodule dataset: (a) WTAM-YOLO; (b) YOLO-Bi_FPN; (c) YOLO-SwinTransformer; (d) YOLO-EfficientNetV2.

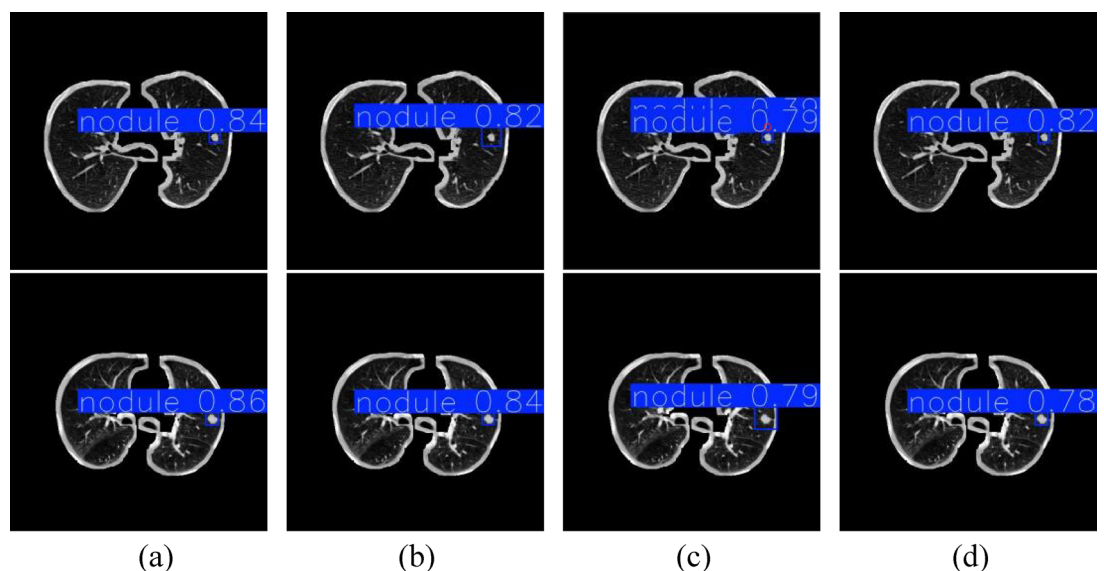


Fig. 14. Prediction results produced by different models on the LUNA16 dataset: (a) WTAM-YOLO; (b) YOLO-Bi_FPN; (c) YOLO-SwinTransformer; (d) YOLO-EfficientNetV2.

achieved a confidence score of 0.84, whereas YOLO-Bi_FPN and YOLO-EfficientNetV2 both reached values of 0.82, and YOLO-SwinTransformer again produced false detections. This further confirmed its limitations in capturing fine-grained features, leading to the misclassification of nontarget objects. Similarly, the second image in Fig. 14 shows that WTAM-YOLO achieved a confidence score of 0.86, outperforming the other models and further validating its superiority in terms of detecting tiny nodules.

In summary, the proposed WTAM-YOLO model exhibited significant advantages when detecting tiny nodules. Its high-confidence detection results verified its ability to capture fine-grained nodule features, thereby providing strong support for enhancing the clinical diagnostic efficacy of lung nodule detection processes.

Conclusions

In the present study, an improved lung nodule detection model, WTAM-YOLO, which is based on YOLOv11, is proposed to address the missed and false detections that are commonly observed in conventional object detection models when addressing small targets. First, WTConv wavelet convolution was employed to reconstruct the

C3k2 module in the backbone network, thereby enlarging the receptive field of the convolution and capturing a broader range of contextual information. Next, the CBAM, which integrates channel and spatial attention, was introduced to enhance the representations of the target features and achieve more precise lesion area localization. In addition, three HRAMi modules were incorporated into the neck network to prevent the loss of fine-grained details during the multilayer feature transmission procedure, particularly for small targets. Finally, the iEMA module was integrated into the high-resolution feature layer of the neck network, combining the strengths of the iRMB and EMA modules to improve the extraction efficiency and recognition accuracy attained for small target features. These improvements substantially enhanced the small targets detection performance of WTAM-YOLO, enabling superior performance to be achieved in lung nodule detection tasks. Compared with the baseline model, WTAM-YOLO achieved improvements of 4.4%, 2.3%, 3.4%, and 9.5% in precision, recall, mAP@50, and mAP@75, respectively, on the lung nodule dataset provided by the Roboflow platform and improvements of 0.7%, 4.6%, 2.5%, and 7.0%, respectively, on the LUNA16 dataset.

Although this study demonstrates significant performance improvements and was validated on publicly available lung nodule datasets, showing stable and superior detection capabilities across different datasets, there is still room for optimization in terms of model size and computational complexity. Experimental results show that the increases in parameters and computational cost of the current model are within a controllable range, and the model demonstrates favorable operational efficiency under existing hardware conditions, providing a foundation for real-time applications in clinical scenarios. Additionally, the method used in this study relies on two-dimensional CT slices for detection, which ignores the three-dimensional contextual information between slices. This may affect the ability to capture the spatial features of the nodules comprehensively and could potentially reduce detection precision in more complex cases. In real clinical environments, factors such as differences in scanning equipment, variations in imaging parameters, and annotation subjectivity may pose challenges to the generalization ability of the model. The real-time performance of the model on standard hardware, along with the interpretability of its outputs for building clinical trust, also warrant attention in the context of clinical deployment. Given the current performance on the pulmonary nodule dataset provided by the Roboflow platform and the LUNA16 dataset, the model can still maintain a high level of focus and accurate identification for the core areas of nodules in clinical data and provide reliable auxiliary detection support for lung nodule screening. Future research could explore multislice fusion or pseudo-three-dimensional input strategies to enhance the ability to perceive the volumetric structure of lung nodules while maintaining its performance advantages and also integrate lightweight design to achieve more efficient clinical deployment and application.

Data availability

The LUNA16 dataset used in this paper is available from website1 (<https://luna16.grand-challenge.org>) and the LungCT-Roboflow dataset is available from website2 (<https://universe.roboflow.com/water-meter-k29v4/lungct-ttywmh>).

Code availability

The source code and trained models are publicly available under the MIT license at <https://github.com/Z-YOLO/O11/WTAM-YOLO/releases/tag/v1.0.0>.

Received: 27 November 2025; Accepted: 24 February 2026

Published online: 27 March 2026

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Author contributions

Yihua Lan and Yi Zhang were primarily responsible for writing the main text of the manuscript and preparing all figures and tables. Yihua Lan focused on overall conceptualization, technical route design, and critical revisions, while Yi Zhang was responsible for the implementation of the models and the execution of the experiments. Jiashu Xu provided important guidance in study design, result analysis, and manuscript preparation, and offered key suggestions for improving the research direction and content. Yingqi Zhang and Tianjiao Hu participated in reviewing and revising the manuscript and supported data analysis. All authors contributed to the study design and discussion of the results, and approved the final version of the manuscript.

Funding

This work is partially supported by the Natural Science Foundation of Henan Province of 2025 (No.252300420141), the 2025 Nanyang Normal University National Natural Science Foundation Cultivation Fund Project (No.2025PY027), the 2025 International Science and Technology Cooperation Project of Henan Province (No.252102520053), the Key Scientific Research Project of Higher Education Institutions in Henan Province (No.25B520011), the Postgraduate Education Reform and Quality Improvement Project of Henan Province (No.YJS2026YBGZZ46), and the Innovation Fund Project for Postgraduates of Nanyang Normal University (No. YJSCX2025037).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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